

Marco Maida

Senior Machine Learning Engineer

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I'm a software and machine learning engineer with over **ten years of experience** delivering systems across **industrial software**, **videogames**, **real-time systems**, **formal verification**, and **autonomous vehicles**.

At **Wayve**, I develop real-time inference systems for autonomous vehicles. My work covers model and runtime co-design, heterogeneous accelerators, numerical validation, and latency optimisation. I work primarily in **C++**, **Rust**, and **Python**, with an emphasis on measurable performance, correctness, and production constraints.

Professional Experience

Wayve Ltd

London, United Kingdom

- **Senior Machine Learning Engineer**

May 2026 → Present

Co-design models and runtime architectures for real-time execution on resource-constrained automotive accelerators. Partition workloads across CPUs, GPUs, and NPUs; start model components as their sensor inputs become available; and maintain validation suites for numerical precision. Optimise end-to-end latency, throughput, and resource utilisation. (*C++*, *Python*).

- **Tech Lead, Inference Performance**

Dec 2024 → May 2026

Led a team of four responsible for Wayve's on-board inference stack. Extended the platform across multiple inference frameworks and accelerators while supporting the same execution path in every vehicle and simulation environment. Owned latency, reliability, and technical delivery for the core inference component. (*C++*, *Rust*, *Python*).

- **Senior Software Engineer**

Dec 2022 → Dec 2024

Joined a seven-person team building Wayve's second-generation on-board platform and shipping it to its first two vehicles. Built the initial real-time inference pipeline; Lidar, GNSS, and microphone stacks; data collection and upload features; and the vehicles' internal software network. (*ROS2*, *Linux*, *C++*, *Rust*, *Python*).

34BigThings

Turin, Italy

Software Engineer – Videogames

Aug 2016 → Sep 2019

Shipped five single-player and online multiplayer titles across Steam, PS4, Xbox One, Nintendo Switch, and mobile. Developed game infrastructure, AI, gameplay systems, and production tools in Unity and Unreal Engine. (*Unity3D*, *C#*, *Unreal Engine*, *C++*).

Maserati / Choralia

Turin, Italy

Software Engineer (Contractor) – Games and Simulation

Sep 2016 → May 2017

Delivered two B2B projects: a browser and mobile educational game, and a 3D product-visualisation tool. Led implementation and supervised an external artist. (*Unity3D*, *C#*, *JavaScript*).

R.O. srl

Software Engineer – Industrial Software

Jun 2013 → Aug 2016

Developed software for glass-processing factories covering order tracking, machine scheduling, waste reduction, and logistics. Managed four junior engineers during the third year. (*C*, *C++*, *C#*, *SQL*).

Education and Research

Research Internship

Jun 2022 → Aug 2022

Bloomberg LP

I worked on accelerating SAT solving using GPUs. (C++, CUDA).

PhD Student

Sep 2019 → Oct 2022

Max Planck Institute

My research focused on formal verification and real-time systems, specifically verifying the timeliness of software systems. I published three papers and mentored three interns. (COQ, C, Rust).

Master's in Computer Science

Nov 2019 → Jul 2022

Technische Universität Kaiserslautern

I specialized in real-time systems as part of a joint master's and PhD program.

Bachelor's in Computer Science

Sep 2016 → Jun 2019

Università degli Studi di Torino

I specialized in computability and formal methods.

Selected Publications

Claycode: Stylable and Deformable 2D Scannable Codes

SIGGRAPH 2025

2025

★ *Journal publication in Transactions of Graphics. Selected for CAF trailer.*

We introduced Claycodes, a new type of visual code that breaks free from traditional rigid grids like QR codes. Claycodes encode data as a tree structure and allow full customizability and deformation, with real-time scanning capabilities.

(<https://arxiv.org/abs/2505.08666>).

From Intuition to Coq: A Case Study in Verified Response-Time Analysis, FIFO Scheduling

RTSS 2022

2022

We developed a formally verified response-time analysis for FIFO schedulers, challenging traditional pen-and-paper methods.

(<https://people.mpi-sws.org/~kbedarka/rtss22.pdf>).

Foundational Response-Time Analysis as Explainable Evidence of Timeliness

ECRTS 2022

2021

★ *Outstanding Paper Award.*

I developed POET, a tool for formally verified worst-case scenario timing analysis.

(<https://drops.dagstuhl.de/opus/volltexte/2022/16336/pdf/LIPics-ECRTS-2022-19.pdf>).

Selected Open-Source Projects

Claycode

<https://claycode.io>

2025

Inventor and lead contributor. Claycode includes a generator website, an Android scanner app, and a large amount of research code.

PROSA

<https://gitlab.mpi-sws.org/RT-PROOFS/rt-proofs>

2020 → 2022

Main contributor to PROSA, one of the leading formally-verified frameworks in the real-time systems community.

Treecode

www.maida.me/treecode

2022

The precursor to Claycode, a novel 2D scannable code that encodes messages as unique trees.

POET

<https://gitlab.mpi-sws.org/RT-PROOFS/POET>

2021

First-ever implementation of a foundational response-time analysis tool, created as part of my first academic publication.

Fast Mobile Cycle (FMC) Framework and Toolkit

www.github.com/340openThings

2018

Developed a Unity3D framework that accelerates production-ready casual game creation, paired with a Python toolkit for bulk operations.